
Social Media and Network Analysis
COSC 2671/ COSC 3047
Assignment 2

Assessment Type	Group assignment. Submit online via Canvas → Assignments → Assignment 2. Marks awarded for meeting requirements as closely as possible. Clarifications/updates may be made via announcements/relevant discussion forums.
Due Date	May 22, 2026, 11:59pm AEST (Week 11, Friday)
Marks	40

1 Overview

In the previous assignment, you had an opportunity to practice using topic modelling and sentiment analysis to perform analysis on Reddit. In this assignment, you'll build upon those skills and work in a team to solve/answer an interesting problem, analysis or question of your choice using social media data.

2 Learning Outcomes

This assessment relates to the learning outcomes of the course which are:

- CLO 1: Apply data science to analyse social media and social networks;
- CLO 2: Analyse social networks by finding communities, identifying important nodes, and influence propagation;
- CLO 3: Analyse social media by applying Natural Language Processing (NLP) techniques to detect sentiment and topics;
- CLO 5: Synthesise and present insights from the social media and network analysis performed.

3 Assignment Details

The goals of the master programs and of RMIT graduates in general are the ability to problem solve, to work in teams and to be able to communicate. This assignment aims to contribute towards your development in these crucial areas. It is essentially a cross between a hackathon and a final year project, where you work on a problem to solve or answer in a team, and in a length of time that is between a hackathon and a final year project.

The assignment can be broken up into a few parts, corresponding to how you could approach it (and also corresponding with the marking rubric). The parts are as follows, and in the following we'll provide further details of each:

- Team formation
- Problem/Question construction
- Team management & Mentoring
- Perform the analysis/Solve the problem
- Communicating the results or analysis

3.1 Team Formation

This assignment is a team assignment. Solo submissions are not permitted. Teams should normally consist of 3 students (2 is also allowed). While groups of two are allowed, we strongly recommend forming groups of 3, as this generally allows for better coordination and a more balanced contribution from all members.

This allows the scope to be larger, so more interesting problems or analyses can be done, as well as allowing you to support each other in a team. The initial task of the assignment is to form the teams. We can help you with this if you are having problems, but our strong suggestion is to select team members with a similar approach to assignments and study. Ideally, you also want a team with a good mix of skills, e.g., analytics, programming/data collection, communication etc.

Teams of 2 and 3 will be assessed using the same marking rubric and must meet the same minimum requirements. However, each team should choose a project scope that is realistic for the number of members in the team.

Teams must consist only of students enrolled in the **same** course code. COSC 2671 and COSC 3047 students may not form mixed teams.

Once you have confirmed your team, go to Canvas. We have set up groups for Assignment 2, where you can self-assign to one of the teams. Please only enter a group allocation once you have formed the team. Please do not put yourself in a team in the hope that others will join you; use the appropriate discussion forum first to form a team.

Each team could nominate a team leader. The team leader is responsible for coordinating communication with the teaching team and coordinating the final submission. Being the team leader does not mean that the student is expected to complete more work than other members.

3.2 Problem or Question Construction

You will also need to select the problem, analysis aim, or research question your team will work on. Your project may be problem-solving or research-oriented. Exploratory analysis is allowed, but it must still have a clear guiding aim and explain what would count as useful or convincing findings.

Each team must submit a topic proposal by **Friday 8 May 2026, 11:59 pm AEST**. The teaching team will provide the detailed topic proposal and consultation process on Canvas.

Your topic proposal and final report should make clear:

- what problem, question, or exploratory aim the team is investigating;
- why it is worth investigating using social media and network analysis;

- what data source or type of data the team expects to use;
- what network will be constructed or analysed, including what the nodes and edges represent;
- what success criteria will be used to judge whether the analysis has answered the question, solved the problem, or produced useful exploratory insights.

As a guide about the scope of problems that are doable, consider the following:

- Beyond statistics, who is the best and most influential cricketer?
- Who is the centre of the Marvel/Nintendo universe? Can we promote Browser to be the most central Nintendo head honcho?
- Github & communities - all things about community analysis on Github.
- Donald vs Kim - who has the bigger ego?
- Social Media Neighbourhood Watch - Alerts for where, when and what crimes are been committed in Melbourne?
- What is trending in the area of Climate change on social media?
- What are the characteristics of successful and popular designs on Canva?

If you look at the problems, some are more analysis/research in nature, others are about solving a problem. This is not an exhaustive list. There is no bias in assessment towards the different types, as we know each of you has different backgrounds, knowledge, and interest, so would like to leave it open to you. There are some constraints that your assignment should adhere to, though:

- Must include at least one source of social media.
- Must include a graph or network and its analysis/processing to solve a problem.
- [COSC 2671 Postgraduate Students Only] Must include network structure & influence analysis, for instance, you could consider - How do different network measures (e.g., centrality, clustering, community structure) reveal the roles of users in a social network, and how can these insights be used to identify influential nodes or communities?
- Must include some element of analysis using social media and networks, and something we learnt in class. It cannot be all machine learning, for example.
- Your analysis or solution is to be implemented primarily in Python. R is not permitted, including for visualisation submitted for assessment. Other languages may be used only for optional supporting components, such as a prototype front end.
- All code should be your own; you can leverage packages such as Networkx, etc., but it shouldn't be copied from an existing solution.

If unsure, please contact us.

3.3 Data Requirements

Your dataset must be suitable for both graph/network analysis and NLP or text-based analysis. This means the data should contain enough relational information to construct or analyse a network, and enough textual content to support meaningful NLP, text, topic, sentiment, or similar social media analysis.

Acceptable data sources include social media platforms, online social platforms, public forums, online communities, public repositories, public datasets, or APIs, provided the data is appropriate for the project question and complies with relevant platform terms and legal or ethical restrictions. Examples may include Reddit, X/Twitter, YouTube comments, GitHub, public forums, or public benchmark datasets.

You may collect data yourself or use an existing public dataset. In either case, the report must clearly state:

- where the data came from;
- how it was collected or obtained;
- the collection date or time period covered, where relevant;
- the main fields or variables used in the analysis;
- how the data supports the network analysis component;
- how the data supports the NLP or text analysis component;
- any important limitations, missing data, sampling issues, or access restrictions.

Do not submit private, sensitive, confidential, or credential data. Do not include API keys, access tokens, passwords, or private account information in the submitted files. If the full dataset cannot be submitted because of size, privacy, copyright, platform terms, or access restrictions, submit a representative sample that shows the data structure and explain the limitation in the report or a README file.

3.4 Network Analysis Requirements

All projects must include a meaningful network analysis component. A network visualisation alone is not sufficient. Your report should explain how the network was constructed and how the network analysis helps answer the project question or aim.

At minimum, the report should state:

- what the nodes represent;
- what the edges represent;
- whether the network is directed or undirected, and why;
- whether the network is weighted or unweighted, and why;
- any filtering, thresholding, sampling, or preprocessing decisions used to construct the network;
- at least one relevant network measure, algorithm, or analytical method used to interpret the network.

The network analysis should be connected to the project aim. For example, if the project is about influential users, the report should explain why the chosen influence or centrality measure is appropriate. If the project is about communities, the report should explain how communities are detected and how they are interpreted.

For COSC 2671 projects, network structure and influence analysis must go beyond listing a centrality ranking. The report should interpret what the network structure or influence results reveal about user roles, communities, interaction patterns, information flow, or another relevant aspect of the project question.

3.5 Team Management & Mentoring

Although it isn't part of this course, we know from our previous experiences as students and lecturers that there are potential difficulties when working in a team. Some of you are very experienced in working as a team, others may do with a few pointers, so some of the following is advice and suggestions that can help with your team dynamics and management:

- Each team is different and what works best is team dependent, but the most successful teams usually have either all highly motivated and self-disciplined members or a team leader who ensures the teams don't lose sight of targets and drive the team towards them. Regardless of style, we strongly suggest to nominate a leader, and to agree upon vesting some authority with them, e.g., keep everyone on track etc.
- Set up regular meetings within your team, whether in person, on Skype or Slack. Have an agenda for each meeting, and a plan for the assignment.
- Communicate with each other and try to stick to timelines/plans. If can't stick to them, have remedial actions.
- Once you have someone as the team leader, please specify within the canvas groups. We'll mainly liaison with the leaders (from our end) and the leader can disseminate the information to the team members.
- Use timesheets and other means to record progress, which is part of your submissions and to avoid arguments later about contributions, and setup a code repository (e.g., Github or Bitbucket), a MS Teams team and/or Google drive to easily share code. The repository/shared drive is a good way to keep everything together and also to ensure progress is made. In the event there is an argument about contributions, it is also a way for us to resolve who did what.

3.6 Perform the analysis/Solve the problem

The next component is to gather or obtain the necessary data, perform the analysis, and iterate on your approach. You are encouraged to use tools and techniques studied in this course, and you may also use relevant techniques from other courses, such as data visualisation, time series analysis, or machine learning, where these support the project aim.

The core data processing, network construction, NLP/text analysis, modelling, and assessed visualisation must be implemented primarily in Python.

You may use standard Python packages, such as pandas, NumPy, scikit-learn, NetworkX, NLTK, spaCy, matplotlib, seaborn, Plotly, or similar libraries. Package use should be appropriate and acknowledged where relevant. You must not copy an existing end-to-end solution.

Your report should explain why you selected each major analytical approach and how it helps answer the project question or aim. Exploratory work is encouraged, but the final report should focus on the methods and results that are most relevant to the project.

3.7 Communication

The final component is to communicate your solution, analysis and discussions. This is done in two ways. One is via a report, much like what you did for assignment 1. The second is a group presentation, where we want each group to present and “pitch” their problem and analysis during the last week of the semester. Each member must speak in the presentation.

4 Submission

- Your report in PDF, named `Report_student_number_PG/UG_Group_number.pdf` (e.g., `Report_s1234567_UG_Group_89.pdf`), up to 20 A4 pages in length and in font size 12 of assessed content, not including appendices. Note this is a maximum, not a length you must have. Anything beyond 20 pages will not be read, and anything in the appendix should be considered as additional information, as there is no guarantee it will be read. Please do not ask for more pages; this limit is strict.
- Your code files, submitted as notebooks, Python scripts, or a compressed code folder, named clearly using your student number, course level, and group number (e.g., `student_number_PG/UG_group_number.ipynb`). The submitted code should include the scripts or notebooks used to produce the analysis and results reported. The code should be readable, commented where appropriate, and organised so that the teaching team can understand the workflow.

Where relevant, include a short README or instructions describing the order in which files should be run, required packages or dependencies, and expected inputs. The submitted code should be consistent with the results in the report. Do not include API keys, access tokens, passwords, or private credentials.

- Access demonstration file, named as `Access_student_number_PG/UG_group_number.txt`, share access with us to your code and/or document shared repository or drive. Include the details how we can access it.
- Worksheet file in PDF, named `Worksheet_student_number_PG/UG_group_number.pdf`. The worksheet must include the team project plan, weekly timesheets for each team member, and individual self-reflection. Each team member must maintain a weekly timesheet. The self-reflection section must be no more than 1 A4 page in total, font size 12.
- A representative sample of the data, no more than 10 MB in size. The sample should be sufficient to show the structure of the data used for the network and

NLP/text analysis. If the full dataset cannot be shared, explain why in the report or in a README file submitted with the sample.

Submission should be made to Canvas. Closer to submission we will describe the submission process in more detail.

5 Assessment

The assessment is based on your presentation, report and your progress.

Examine the assessment rubric below.

You'll be assessed on how well you communicated your work, on the problem selected, how your team went about the assignment and your insights/solution. As a rough rule of thumb, you can get a credit and distinction if you just use the code we learnt in class, but to get an HD would require you to do some research and go beyond what we did in class.

In addition, part of your assessment would be self-reflection on your own performance and your team's performance. Here you may add any additional information you'll like to inform us of difficulties. The worksheet provides evidence for the Teamwork and Management criterion, including planning and progress tracking. We'd also like you to provide an estimate of the work breakdown (via timesheets) and your assessment of contributions in the self-reflection, and to ensure an even work distribution if possible, it will be up to our discretion for marks redistribution if the workload is unreasonably unbalanced.

6 Academic integrity and plagiarism (standard warning)

Academic integrity is about honest presentation of your academic work. It means acknowledging the work of others while developing your own insights, knowledge and ideas. You should take extreme care that you have:

- Acknowledged words, data, diagrams, models, frameworks and/or ideas of others you have quoted (i.e. directly copied), summarised, paraphrased, discussed or mentioned in your assessment through the appropriate referencing methods
- Provided a reference list of the publication details so your reader can locate the source if necessary. This includes material taken from Internet sites. If you do not acknowledge the sources of your material, you may be accused of plagiarism because you have passed off the work and ideas of another person without appropriate referencing, as if they were your own.

RMIT University treats plagiarism as a very serious offence constituting misconduct. Plagiarism covers a variety of inappropriate behaviours, including:

- Failure to properly document a source
- Copyright material from the internet or databases
- Collusion between students

For further information on our policies and procedures, please refer to the following:

<https://www.rmit.edu.au/students/my-course/assessment-results/academic-integrity>.

We will run both code and report similarity checks.

7 Getting Help

There are multiple venues to get help. First point of call should be Canvas, assignment FAQ (on Canvas), recordings about it and the discussion forum. There will also be weekly online consultation hours for the assignment (see Canvas for times). Please **refrain from posting solutions** to the discussion forum.

A Rubric (40 marks)

Use the following rubric to help you determine how to approach the assignment. Note that teamwork and management are both a team and individual mark, based on how your team functions, manages itself, and your self-reflections.

Criteria	Excellent	Good	Fair	Poor
Problem Formulation (5 marks)	The proposed problem/question is interesting, well-motivated and non-trivial. Problem and success criteria are well designed and thought out, and the scope is realistic.	The proposed problem/question is interesting, relatively well-motivated, and non-trivial. Problem and success criteria are reasonably designed and thought out, and scope is realistic.	The proposed problem/question is somewhat interesting, has some motivation, and is on the simple/trivial side. Problem formulated for the sake of completing the assignment. Problem and success criteria are not well designed and thought out, and the scope may be unrealistic.	The proposed problem/question is not very interesting, has minimal motivation and on the simple/trivial side. Problem formulated for the sake of completing the assignment. Problem and success criteria are not well designed and thought out, and scope may be unrealistic.
Teamwork and Management (7.5 marks)	Team functions well and has effective management procedures. Have a well-designed project plan and weekly plans. Timesheets are realistically filled in and individual self-reflection is insightful and reflects thought has been given to what worked, what didn't and how to improve in the future.	Team functions well and has reasonable management procedures. Have a decent project plan, weekly plans. Timesheets are realistically filled in and individual self-reflection is reasonably insightful and reflects thought has been given to what worked, what didn't and how to improve in the future.	Team functions more as a collective of individuals and there is some communication demonstrated between members. Management procedures are somewhat lacking. Have a project plan and weekly plans that reads to be done before submission. Timesheets appear manufactured and individual self-reflection is not very insightful and rushed.	Team functions more as a collective of individuals and there is little communication demonstrated between members. Management procedures are lacking. Lacking or minimal project plan and weekly plans that read to be done before submission. Timesheets appear manufactured, and individual self-reflection is not insightful and rushed.
Approach (7.5 marks)	The approach is an appropriate method to take to solve the problem or answer the analytical question. The approach taken goes beyond using the tools provided in class. The team justifies and explains their approach well. The approach includes data collected and techniques used.	The approach is an appropriate method to take to solve the problem or answer the analytical question. The approach taken is limited mostly the tools provided in class. The team justifies and explains their approach well. The approach includes data collected and techniques used.	There are other approaches that are clearly a more appropriate method to take to solve the problem or answer the analytical question. The approach taken is limited to the tools provided in class. Team somewhat justifies and explains their approach, but there are unexplained choices. The approach includes data collected and techniques used.	There are other approaches that are clearly a more appropriate method to take to solve the problem or answer the analytical question. The approach taken is limited to the tools provided in class. The team does not justify or explain their approach. The approach includes data collected and techniques used.
Analysis/result & Discussion (10 marks)	Problem solving: The solution solves the problem well and all the success criteria are satisfied. Team is able to provide analytical and/or empirical evidence of this. Answering analytical question & Analysis component: Excellent discussion of results that answers the question proposed or contributes towards solving the problem. Conclusions are supported by analytical and/or empirical evidence. All success criteria are answered.	Problem solving: The solution solves the problem adequately and most of the success criteria are satisfied. Team is able to provide analytical and/or empirical evidence of this. Answering analytical question & Analysis component: Good discussion of results that answers the question proposed or contributes towards solving the problem. Conclusions are supported by analytical and/or empirical evidence. Most success criteria are answered.	Problem solving: The solution somewhat solves the problem and about half of the success criteria are satisfied. Team provides some analytical and/or empirical evidence of this. Answering analytical question & Analysis component: Adequate discussion of results that answers the question proposed or contributes towards solving the problem. Conclusions are somewhat supported by analytical and/or empirical evidence. About half the success criteria are answered.	Problem solving: The solution does not really solve the problem and many of the success criteria are not satisfied. Team is unable to provide analytical and/or empirical evidence of this. Answering analytical question & Analysis component: Minimal discussion of results that answers the question proposed or contributes towards solving the problem. Conclusions are not supported by analytical and/or empirical evidence. Many of the success criteria are not answered.
Report Presentation (5 marks)	Report is easy to read and flows well. It is structured well, leading the reader through the process of answering the questions or solving the problem. Tables, figures and other visualisation are easy to read and to interpret.	Report is reasonably easy to read and flows relatively well. It is structured reasonably well, leading the reader through the process of answering the questions or solving the problem. Tables, figures and other visualisation are easy to read and to interpret.	Report is difficult to follow in places and doesn't flows well. It is adequately structured, but reader may find it difficult to follow through the process of answering the questions or solving the problem. Tables, figures and other visualisation are either too small or difficult to interpret.	Report is difficult to follow and doesn't flows well. It is bare structured, but reader may find it difficult to follow through the process of answering the questions or solving the problem. Tables, figures and other visualisation are either too small or difficult to interpret.
Group Presentation (5 marks)	Presentation is well structured, clear and easy to follow by master students. It is engaging and interesting and able to capture the audience's attention. A candidate for best presentation.	Presentation is reasonably well structured, clear and easy to follow by master students. It is reasonably engaging and interesting and able to capture the audience's attention.	Presentation has some structure, somewhat clear and easy to follow by master students. It is not that engaging and interesting and unable to capture the audience's attention.	Presentation has little structure, is not clear and not easy to follow by master students. It makes little attempt to be engaging and interesting and unable to capture the audience's attention.